

University of San Francisco

Physics 303/CS 486/CS686 – Deep Learning & Bayesian Learning

Homework 4

11 PM, Friday, April 7, 2026

**Cats and Dogs**

***You need to be in class in person in order to compete Part V***

### Part I. Downloading the image data

<https://www.microsoft.com/en-us/download/details.aspx?id=54765>

### Part II. Preprocessing

A) Make sure that each image has 3 channels (RGB). Sometimes you encounter images with 4 channels. In that case, it's almost always the case that one of the channel is the transparency channel. RGBA being one of the most common pattern for images that include transparency. In the RGBA model, A (Alpha): Represents the transparency of the pixels. This channel controls the transparency of the image. For example, for an 8-bit image, a value of 0 indicates full transparency (completely clear) and a value of 255 indicates full opacity (completely solid). There are also one or two images that are corrupted (0 KB), one of which has an name of "666". When you submit your homework, please include code to remove these images, as well as to ensure that all images have only 3 channels.

B) Normalize pixels as you saw in class. Sometimes they are rescale to 0 and 1, or  $-1$  and 1, or 0 and 255. This can depend on the architecture you adopt (see next part).

C) Make sure that all the images have the same size. If not, you may use `resize()` from `skimage` (see the example of the image for the astronaut in one of the notebooks).

### Part III. Architecture and Training

You can:

1. Use a custom architecture of your own design, either a CNN or a ResNet (or MLP)
2. Use TensorFlow or PyTorch
3. Use an "off-the-shelf" architecture (e.g., ResNet18)

- a. You can either use a pre-trained model, or
  - b. Train “from scratch”
4. You may train as many models as you would like. But you may only do one of the following:
  - a. Submit one model, your best model, or
  - b. If you want to use multiple models, specify your “voting” scheme
5. Address all deprecation issues — that way your code will last longer!

For training, use an 8:2 split between the training and validation sets. You can try other ratios if you’d like (between 7:3 - 9:1 are all reasonable).

You can experiment with different optimizers, different activation functions, learning rates, and learning rate decay schedules.

#### Part IV. Performance Metrics

Your code should show the loss vs. training epoch figure, the accuracy vs. epoch figure, the ROC curve, and the Precision-Recall curve.

#### Part V. Classification Challenge

On Tuesday, April 7, in class we will do a classification challenge. **You will turn in the code for this challenge.** Your grade for this homework will not depend on your *performance* in the classification challenge. *But your grade WILL depend on the code that you turn in and your PARTICIPATION in this classification challenge.*

At the end of class, you will submit:

- The notebook or code you used to do the classification challenge
- The ROC curve for cats (without using the 5 “other” images)
- The 5x5 panel of the 25 images, with the title for each image containing “label: cat, pred: cat” or “label: cat, pred: dog”, etc.

#### Extra Credit: Multi-class Classification

Use four classes (instead of just classifying dogs vs. cats):

1. One cat only
2. One dog only
3. More than one cat
4. More than one dog

You need to look at each individual image and label with the class targets (1, 2, 3, 4) as shown above.

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#### For Thursday, April 2

You will need to turn in:

1. Your code for this assignment
2. Your saved model (or models; please see the assignment linked above)

If your code includes the extra credit part, please upload the labeled images (1, 2, 3, 4) either directly on Canvas, or if the file size is too big, upload them on your google drive, and share it with Andrew and myself.

#### On Tuesday, April 7

We will do the classification challenge (Part V) in class. You will see the images and participate in this challenge ONLY IF you are in the classroom in person. On that day, you will turn in the code for preprocessing the images in the challenge and the your classifications. We will talk more about this process on Thursday, 4/2, during class.